STORMROUTE

Weather-Aware Truck Routing — Architecture & Risk-Engine Case Study

Python · Django · Django REST Framework · React 18 · TypeScript · Vite · Tailwind CSS · Mapbox GL · Vercel

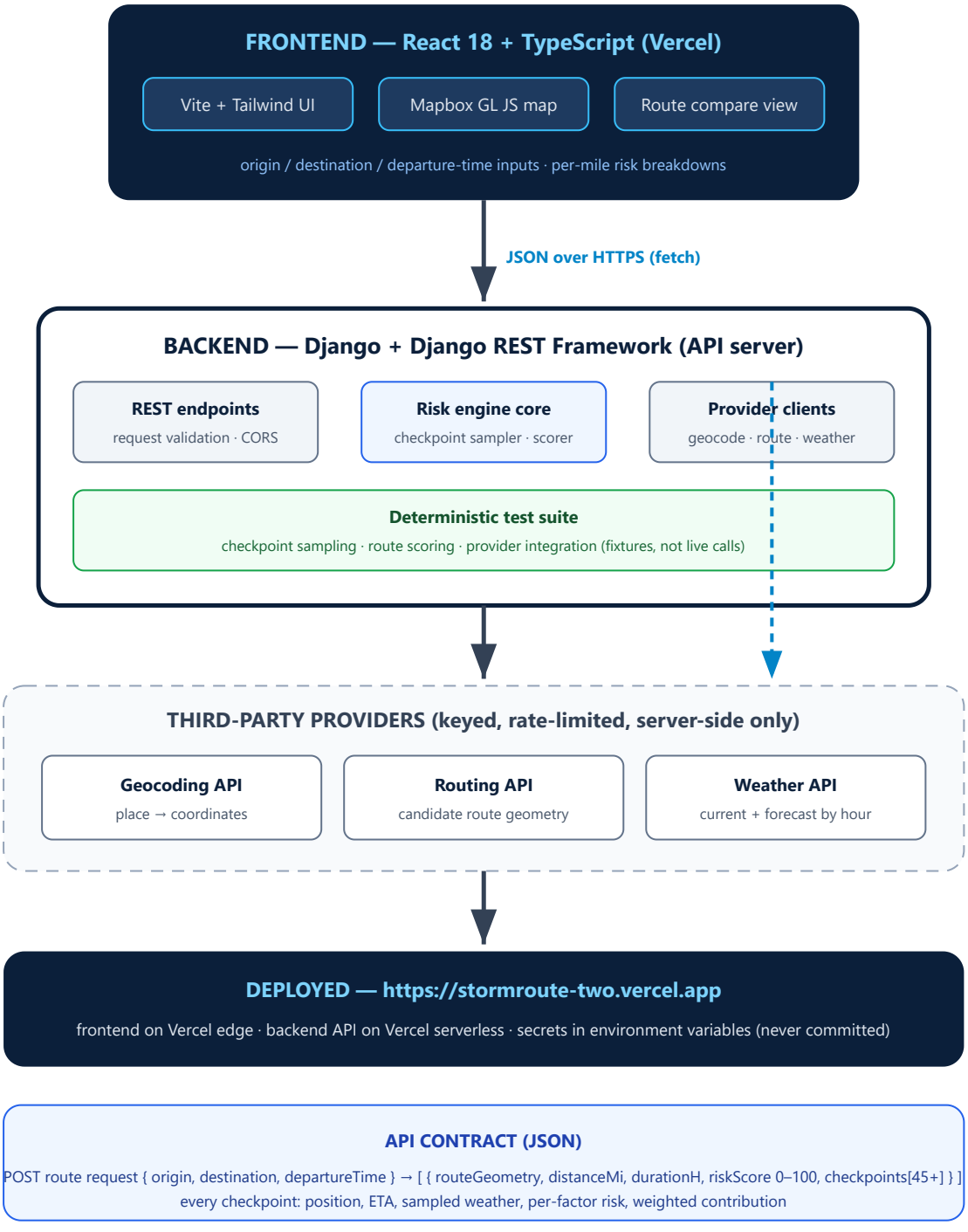
> Enter an origin, destination and departure time — StormRoute samples **45+ weather checkpoints** along every candidate route and ranks each route with a transparent **0–100 per-mile risk score**, so the safest driving option is visible before the wheels move.

PROJECT CASE STUDY · MUHAMMAD SUDAIS · APPLIED AI ENGINEER · OCTOBER 2026

Live app: https://stormroute-two.vercel.app · Video walkthrough: https://www.loom.com/share/c0efe1dd8b684aa1b9fb477ed800a7b6

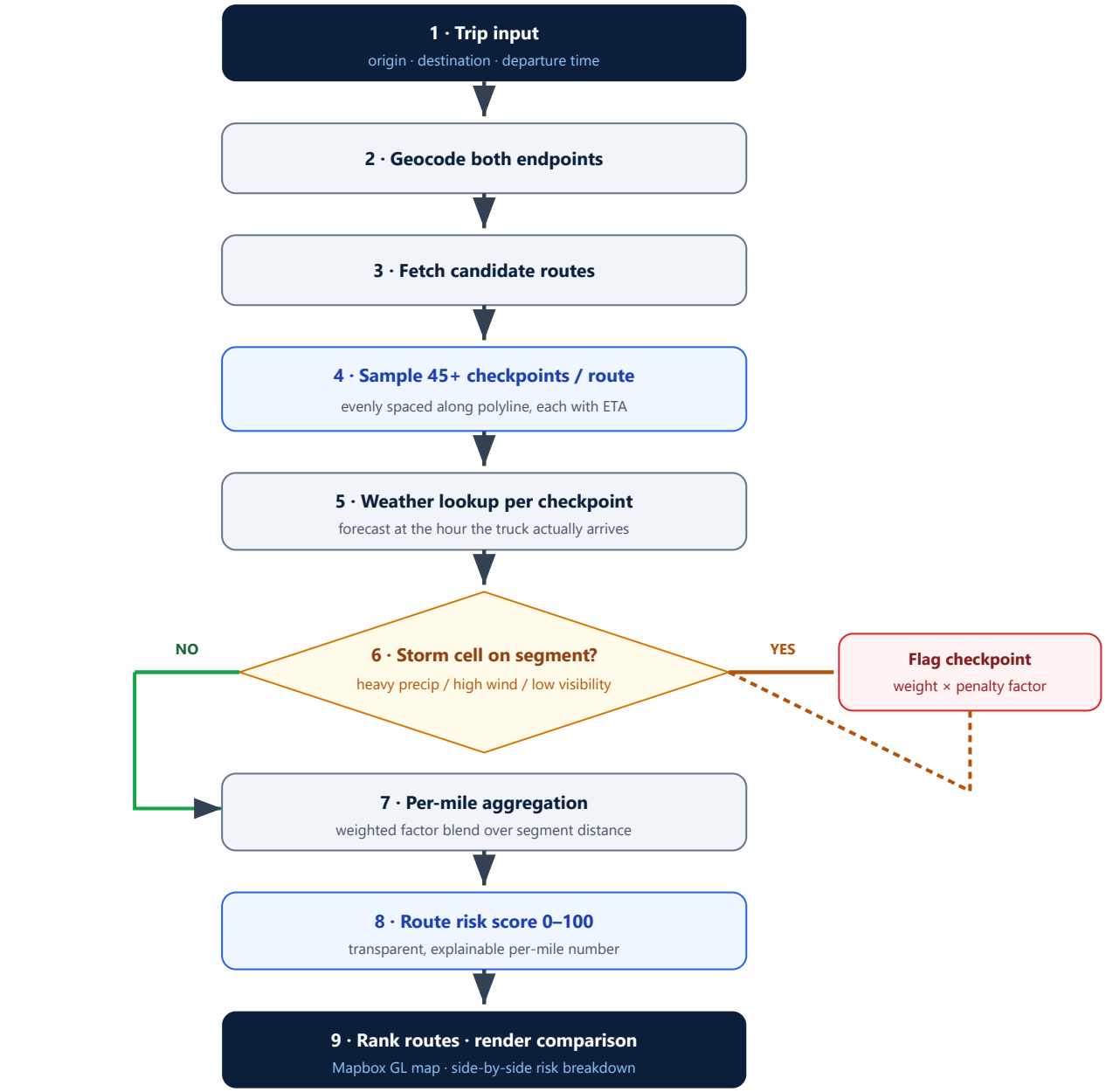
1 · System Architecture

Deployed topology — public frontend, JSON API backend, third-party data providers



2 · Risk-Engine Flow

From trip input to ranked, weather-scored routes



Design rule: no black boxes

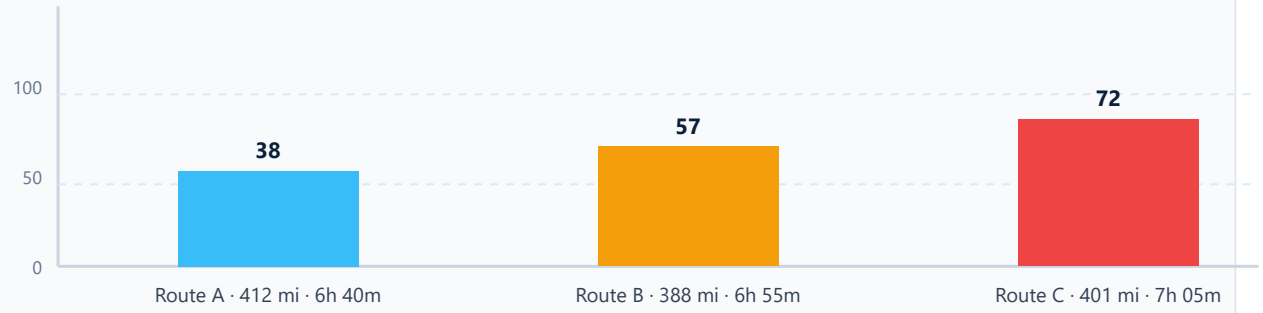
every score can be decomposed into per-checkpoint, per-factor contributions in the UI

3 · Risk Model Outputs

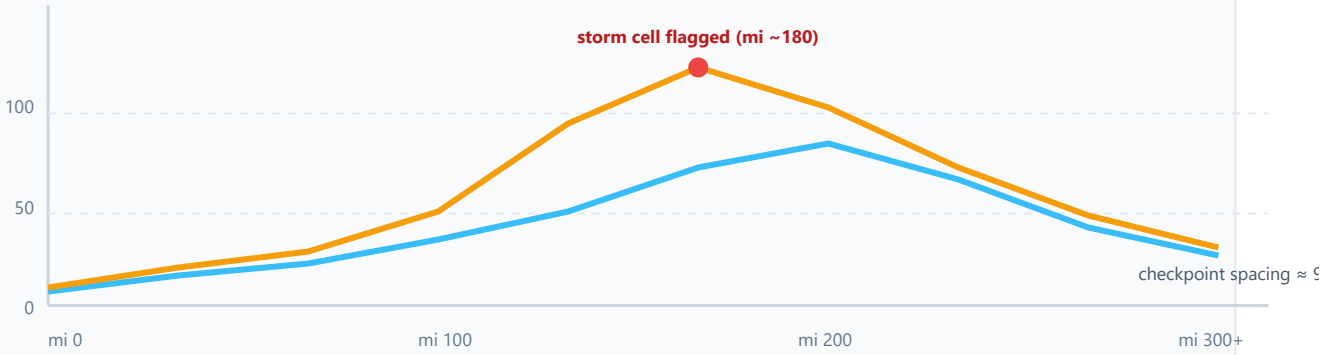
Example engine output for a 3-route plan — illustrative sample, not a live forecast

Route A (recommended) Route B Route C

A · Route comparison — overall risk score (0–100, lower is safer)



B · Per-mile risk profile along the corridor (checkpoint samples)



C · What moves the score — weighted risk factors



- Precipitation (rain / snow intensity) — 34%
- Wind gusts (crosswind on high-profile rigs) — 26%
- Visibility (fog / heavy precipitation) — 18%
- Storm cells flagged on segment — 14%
- Temperature / road-glaze risk — 8%

Note: figures above are an illustrative example of the engine's output format (deterministic sample), not a live forecast. The deployed app computes scores from real provider data at request time.

4 · Stack, Quality & Deployment

What it is built with, how it is tested, where it runs

Technology stack

Layer	Technology	Role
Frontend	React 18, TypeScript, Vite, Tailwind CSS	Trip form, route comparison, risk breakdowns
Map	Mapbox GL JS	Interactive corridor map with flagged checkpoints
Backend	Python, Django, Django REST Framework	JSON route API + risk engine core
Data providers	Geocoding · Routing · Weather APIs	Place resolution, route geometry, forecasts
Hosting	Vercel	Edge frontend + serverless API, CI on push

Quality & testing

Area	Approach
Risk engine	Deterministic Python tests — sampling math, scoring weights, edge cases
Provider integration	Contract tests against recorded fixtures (no live calls in CI)
Frontend	TypeScript strict mode + production build gate
Browser verification	Real-browser E2E passes of trip planning, comparison and map flows

Deployment

Component	Status
Frontend	Live — https://stormroute-two.vercel.app
Backend API	Live — serverless, health-checked endpoint
Walkthrough	4-minute video — https://www.loom.com/share/c0efe1dd8b684aa1b9fb477ed800a7b6

Built by

Muhammad Sudais — Applied AI Engineer (freelance, full-stack). Designed, built, tested and deployed end-to-end as a solo project: product thinking, backend risk engine, frontend dashboard, provider integration, testing and public deployment.

Risk scores are checkpoint-based estimates computed from third-party weather data at request time; they support trip planning and do not replace driver judgment or official road advisories.